

Scalability & Future Plan

Date	10 July 2026
Team ID	XXXXXX
Project Name	Smart Lender AI – AI-Based Loan Eligibility Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document presents the current limitations, scalability strategy, and future enhancements of the Smart Lender AI system. The application successfully predicts loan eligibility using machine learning and provides applicant profile management, prediction history, AI-based analysis, and PDF report generation. The modular architecture allows future integration with cloud services, databases, and real-time financial systems.

Current System Limitations:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	Application Deployment	Runs locally using the Flask development server.	Deploy the application using Gunicorn or uWSGI behind an Nginx web server on a cloud platform such as AWS, Azure, or Render for improved scalability and availability.
2	Database Management	No persistent database is used.	Integrate PostgreSQL or MySQL to securely store user accounts, loan applications, prediction history, and system logs.
3	Batch Processing Performance	Batch prediction processes uploaded CSV files sequentially.	Implement asynchronous task processing using Celery with Redis to improve performance when handling large datasets.
4	Security	Basic input validation is implemented without authentication or API protection.	Introduce secure user authentication, role-based access control, encrypted credentials, environment variables, HTTPS, and API rate limiting.
5	Machine Learning Model Management	The model must be retrained manually whenever new training data becomes available.	Develop an automated model retraining pipeline capable of periodically updating the deployed model using newly collected loan application data.

Scalability Plan:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local Development Environment	The application currently runs on the Flask development server (<code>localhost</code>) and is intended for demonstration purposes only. It	High

		cannot efficiently handle multiple concurrent users or production-level traffic.	
2	No Persistent Database	User information and prediction history are not stored permanently. Once the application is restarted, previous prediction records are unavailable.	High
3	Limited User Authentication	The current system does not include user registration, login, or role-based access control, allowing unrestricted access to the application.	High
4	Static Training Dataset	The machine learning model is trained on a fixed historical dataset and does not automatically learn from newly submitted loan applications.	Medium
5	Local File-Based Model Storage	The trained model and preprocessing artifacts are stored as local files (.pkl), making collaborative deployment and centralized model management difficult.	Medium

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Database Integration and User Authentication	Week 2	Integrate a relational database and implement secure user authentication to support persistent data storage, user management, and application history.
Phase 2	\ Cloud Deployment and Continuous Integration	Week 3	Containerize the application using Docker and deploy it to a cloud platform with automated testing and continuous integration pipelines for reliable deployment.
Phase 3	Third-Party Financial API Integration	Week 4	Integrate external credit bureau APIs and financial services to retrieve verified credit scores and financial information automatically, improving prediction reliability.
Phase 4	Advanced Explainable AI and Model Management	Week 5-6	Enhance prediction transparency using SHAP or LIME explanations and develop an administrative dashboard for automated model retraining, monitoring, and performance management.